



GE INDUSTRIAL MOTORS

a **WOLONG** company

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Customer:

DataPack Document, Version: v23095047.01

MODEL NUMBER: 5KAF513SAA151

Outline Drawing: 239C6C00AAK

Connection Diagram: GEM2034E-FIG23

Instruction Book: GEI-56128

Design Code: 50BD0299A

Type: KAF

Frame: 5013ST

Phases: 3

Poles: 2

Output Power: 800HP 596.6KW

RPM: 3580

Voltage: 460

Hertz: 60

Amps - FL: 828.0

Service Factor: 1.00

Time Rating: CONT

Estimated Weight: 6917 Lbs

Enclosure: TEFC

Encl Construction: SD

Ambient Max(°C): 40

Insulation Class: F

NEMA Design: --

Nominal Efficiency: 96.2%

KVA Code: J

Max KVAR: 123.4

Power Factor: 94.0

Bearing - DE: 6315ZC3 VL0241

Bearing - ODE: 6315ZC3 VL0241

Vibration: 0.1 IN/SEC

Bearing Lubrication: MOBIL POLYREX EM

Protection(IP): IP 55

Enclosure is Totally Enclosed Fan-Cooled

Stamped Nameplate Notes:

TEMP CONT HTR LDS HE1-HE2 115V 200W
ROT CW FACING ODE LEAD/PH SEQUENCE 1-2-3/1-2-3
MAX SPACE HEATER SURFACE TEMPERATURE 208C
STAMP NP249A5564P075 AS BELOW:
MODEL:5KAF513SAA151
SN: XXX CSA CERTIFIED CSA.09.2216219
FOR CL I DIV2 GRP A,B,C,D T2B IN -40C<= TAMB<= 40C
1.0SF ON SINE-WAVE PWR OR ALTERNATE RATING
TEMP CODE -- AT -- SF ON SINE-WAVE PWR
OR T2B VT PWM CONTROL 1.0 SF 40 C AMB VT 0-60 HZ.
FOR DIRECT COUPLED LOAD ONLY
CONSTANT HP RANGE 60-63 HZ
2P - ST EXTN

Additional Information:

C/BOX 5700 CU IN - 3(3.00" NPT)
C/B GRD PLATE
100 OHM WINDING RTD LEADS TO AUX BOX AT F2 DE
SUGGESTED WINDING RTD SETTINGS
ALARM 155C TRIP 165C

Performance Characteristics

1st Winding 1st Connection

Design: 50BD0299A

LOAD %	125.0	115.0	100.0	75.0	50.0	25.0	0.0
% EFF	96.81	96.91	96.20	97.09	96.79	95.08	0.00
% PF	94.55	94.46	94.00	92.77	88.70	73.51	4.71
AMPS	1022.54	940.63	828.00	623.48	436.11	267.84	172.05

TORQ(FL)#FT 1173.75

TORQ(LR)%FL 158.11

TORQ(BD)%FL 431.04

AMPS(LR) 7649.38

PF AT START 0.17

This motor is capable of two cold or one hot start with a maximum connected load inertia of 2223.0 Lb-Ft Sq (93.7 Kg-meter Sq) at 100% voltage, where the load torque varies with the square of the speed. Acceleration time with maximum inertia and the above load type is 17 seconds. Safe stall time at 100% voltage is 34 seconds cold, 20 seconds hot. Rotor inertia is 156.68 Lb-Ft Sq (6.603 Kg-meter Sq).

Open Circuit A-C: 2.071

Short Circuit D-C: 0.042

Short Circuit A-C: 0.045

X/R Ratio: 15.772

Stator Slots: 48

Rotor Slots: 40

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Speed Torque Current Curve (First Connection, First Speed)

